

**Probability
Fall 2026**

**Homework I.
Due September 2, 2026**

If you use Mathematica you must hand in any Mathematica Notebook that you use to complete the assignment. I generally do not want electronic copies; just paper please.

Practice.

Chapter 1, problems 4, 5, 6, 7 and 17. (Do not hand in practice problems.)

Required Homework.

1. Let E and F be events. Using the axioms of probability, prove that

$$P(E \cup F) = P(E) + P(F) - P(EF).$$

And work the following problems from the text:

8. (Bonferroni's inequality), 10. (Boole's inequality), 12, 39 and 41.